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Inventor(s)	Sholl; Sandra et al.

Using machine learning models to generate a mirror representing an image of virtual try-on and styling of an actual user

Abstract

Disclosed are example implementations of systems and methods for virtual try-on of articles of clothing. An example method of virtual try-on of articles of clothing includes ingesting information specifying articles of clothing using a first machine learning model; retrieving a stored outfit, the stored outfit comprising at least one garment; inferring a style for the stored outfit preferred by the user with a second machine learning model; generating a shopper avatar wearing the stored outfit with the inferred style based on a physique of a user; and providing the shopper avatar wearing the stored outfit with the inferred style for display to the user in a user interface.

Inventors: Sholl; Sandra (West Hollywood, CA), Freede; Adam (West Hollywood, CA)

Applicant: Zelig Technology, LLC (Los Angeles, CA)

Family ID: 1000008821029

Assignee: Zelig Technology, LLC (Los Angeles, CA)

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Primary Examiner: Harrison; Chante E

Attorney, Agent or Firm: Patent Law Works LLP

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/378,593, filed Oct. 10, 2023, titled “Systems And Methods For Using Machine Learning Models To Effect Virtual Try-On And Styling On Actual Users”, which is a continuation-in-part of U.S. application Ser. No. 18/217,412, filed Jun. 30, 2023, titled “Systems And Methods For Using Machine Learning Models To Effect Virtual Try-On And Styling On Actual Users” which claims the benefit of U.S. Provisional Application Ser. No. 63/358,038, filed Jul. 1, 2022, titled “System For Using Machine Learning Models To Effect Virtual Try-On And Rental Of Digital Garment Transfers On Actual Users With Selected High Fashion Styling Options And Looks,” all of which are hereby incorporated herein by reference in their entirety.

FIELD

(1) The disclosure relates generally to the field of image processing and, more particularly and not by way of limitation, some implementations are related to image processing systems facilitating online try-on of articles of real-world clothing virtually. Still more particularly, the present

disclosure relates to using machine learning models to generate a mirror representing an image of virtual try-on and styling of an actual user.

BACKGROUND

(2) The ubiquity of desktop and laptop computers, smart phones, tablet computers and other personal computing devices has enabled has made it commonplace to engage in personal and professional online activities on a daily basis. Because many such devices include cameras, users may capture images or video sequences of themselves or others and use the acquired images, either after editing them or otherwise, in social and commercial interactions.

(3) In the commercial realm, there have been recent attempts to leverage personal computing devices including cameras in online shopping activities by enabling users to be virtually dressed in clothing articles available for online purchase. However, many of these systems have limitations which have impeded their widespread deployment. For example, certain systems require a user to provide views of themselves from a number of different perspectives, which may require the assistance of another person and can be tedious and time consuming. Existing systems are also not known to permit users to select a desired styling of a clothing article (e.g., sleeves rolled up, sleeves below wrist, garment tucked in or untucked, etc.) when virtually trying on the article. Moreover, existing try-on systems are generally incapable of manipulating clothing articles to be tried on to realistically drape or deform clothing on a variety of different user poses.

(4) Accordingly, a need exists for an improved way of virtual try-on and styling of clothing on a user.

SUMMARY

(5) In one example implementation, an implementation includes systems and methods for virtual try-on and styling of real-world clothing virtually using computer vision and machine learning techniques.

(6) Disclosed are example implementations of systems and methods for virtual try-on of articles of real-world clothing virtually. An example method of virtual try-on of articles of clothing includes selecting a garment from a pre-existing database. The method also includes loading a photo of a source model wearing the selected garment. Additionally, the method includes generating a semantic segmentation of the model image. The method also includes extracting the selected garment from the photo of the model. Additionally, the method includes determining a correspondence between a target model and the source model by performing a feature point detection and description of the target model and the source model and performing feature matching and correspondence validation. The method also includes performing garment warping and alignment of the extracted garment. Additionally, the method includes overlaying and rendering the garment.

(7) According to one aspect of the subject matter described in this disclosure, a method includes ingesting information specifying articles of clothing using a first machine learning model; retrieving a stored outfit, the stored outfit comprising at least one garment; inferring a style for the stored outfit preferred by a user with a second machine learning model; generating a shopper avatar wearing the stored outfit with the inferred style based on a physique of a user; and providing the shopper avatar wearing the stored outfit with the inferred style for display to the user in a user interface.

(8) In general, another aspect of the subject matter described in this disclosure includes a system comprising one or more processors and memory operably coupled with the one or more processors, wherein the memory stores instructions that, in response to the execution of the instructions by one or more processors, cause the one or more processors to perform the following operations of ingesting information specifying articles of clothing using a first machine learning model; retrieving a stored outfit, the stored outfit comprising at least one garment; inferring a style for the stored outfit preferred by a user with a second machine learning model; generating a shopper avatar wearing the stored outfit with the inferred style based on a physique of a user; and providing the

shopper avatar wearing the stored outfit with the inferred style for display to the user in a user interface.

(9) Other implementations of one or more of these aspects include corresponding systems, apparatus, and computer programs, configured to perform the actions of the methods, encoded on computer storage devices.

(10) These and other implementations may each optionally include one or more of the following features. For instance, features may also include generating the shopper avatar wearing the stored outfit and providing the shopper avatar wearing the stored outfit for display to the user in the user interface are performed in real-time so that the user interface serves as a mirror through which the user can see what they would look like in real life. For instance, the method further comprises receiving a selection of a new garment by the user, the selection initiating a virtual try-on of the new garment to replace the at least one garment with the new garment based on the new garment and the at least one garment having a same type of clothing; configuring the shopper avatar to wear the new garment according to one or more styling options; and providing the configured shopper avatar wearing the new garment for display to the user in the user interface. For example, the method may also include selecting one or more additional garments of clothing to recommend to the user using a third machine learning model in accordance with the style of the user; updating the user interface to include the one or more additional garments along with the shopper avatar; and providing the updating the user interface for display to the user. In general, other aspects of the subject matter of this disclosure may be implemented in methods where the information specifying articles of clothing is from one or more from a group of: a designer, a retailer, a B2B customer, a website, a B2B application, and a mobile application. For example, features may also include ingesting information specifying articles of clothing using the first machine learning model using artificial intelligence and machine learning algorithms and a rendering engine. Still other implementations the ingesting information specifying articles of clothing using the first machine learning model further comprises selecting a subset of a large dataset of images of garments from a pre-existing database, wherein the subset of the large data set of images are labelled with a specific style; generating a digital taxonomy having one or more labels associated with each image in the subset of the large dataset of images, wherein at least one of the one or more labels corresponds to the specific style; training a style classifier using the digital taxonomy applied to the subset of the large dataset of images of garments; and generating a first dataset based on executing the trained style classifier on the large dataset of images of garments, the first dataset comprising a plurality of images of garments meeting a threshold probability of depicting the specific style. For instance, features may also include wherein the shopper avatar is a rendering of the user or a human model. For instance, the method further comprises presenting a user interface with a plurality of human body types; receiving a selection of one of the plurality of human body types; and presenting the selected one of the plurality of human body types as the physique of the user for the shopper avatar in the user interface along with body dimensions. For example, the receiving the selection of one of the plurality of human body types may also include receiving a photograph or a video of a user; analyzing the photograph or video of the user; and selecting one of the plurality of human body types as the physique of the user based on the analyzing the photograph or video of the user. For example, features may also include wherein the second machine learning infers the style for the stored outfit preferred by the user based upon styling guidance and recommendations for the user based on a current and historical profile and current and historical engagements of the user. Still other implementations include wherein the user interface in a window overlaid on a web page of a web browser showing in the web page of a website of a retailer.

(11) The features and advantages described in the specification are not all-inclusive. In particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in

the specification has been principally selected for readability and instructional purposes and may not have been selected to delineate or circumscribe the disclosed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing summary, as well as the following detailed description, is better understood when read in conjunction with the accompanying drawings. The accompanying drawings, which are incorporated herein and form part of the specification, illustrate a plurality of implementations and, together with the description, further serve to explain the principles involved and to enable a person skilled in the relevant art(s) to make and use the disclosed technologies.

(2) FIG. 1 provides an overview of an exemplary system for virtual try-on and rental of digital garments styled in conformity with a fashion standard using machine learning in accordance with the disclosure.

(3) FIG. 2 is a process flow diagram which illustrates a machine learning data pipeline and a machine learning try-on/styling pipeline in accordance with the disclosure.

(4) FIGS. 3A and 3B respectively illustrate exemplary inputs and outputs of the machine learning data pipeline 210.

(5) FIG. 4 provides a flowchart of an exemplary workflow for building a base webdataset and each specific version for different machine learning models used within the try-on/styling pipeline.

(6) FIG. 5 shows an architecture for an exemplary implementation of a multi-class styles classifier which may be trained to recognize garment styles presented to it during a training process.

(7) FIGS. 6 and 7 provide high-level representations of operation and training of a multi-class styles classifier implemented using the classifier architecture of FIG. 5.

(8) FIGS. 8A and 8B provide an overview of operation of an exemplary machine-learning try-on/styling pipeline in accordance with the disclosure.

(9) FIG. 9 illustrates an exemplary try-on/styling orchestrator in accordance with the disclosure.

(10) FIG. 10 illustratively represents an exemplary try-on/styling workflow in accordance with the disclosure.

(11) FIG. 11 represents a U-Net architecture of a type useful in implementing the segmentation generators utilized in the try-on/styling workflow of FIG. 10.

(12) FIGS. 12A-12D illustrate various exemplary input and ground truth imagery for use with the segmentation generators of FIG. 10.

(13) FIG. 13A depicts a grid utilized by a machine learning model for garment warping.

(14) FIGS. 13B and 13C respectively illustrate ground truth and garment ghost imagery utilized by the model for garment warping.

(15) FIGS. 14A-14C illustrate exemplary inputs for a model for try-on synthesis in accordance with the disclosure.

(16) FIGS. 15A-15O define an exemplary digital style taxonomy including a plurality of styling combinations conforming to a fashion standard.

(17) FIGS. 16A-16J provide examples of options for the various styling segments included within the styling combination categories of FIGS. 15A-15O.

(18) FIG. 17 is a flow diagram illustrating an example of steps within a pipeline in accordance with the systems and methods described herein.

(19) FIG. 18 is a flow diagram illustrating an example of steps and sub-steps within a pipeline in accordance with the systems and methods described herein.

(20) FIGS. 19A-19G are diagrams illustrating example photos of a photo shoot in accordance with the systems and methods described herein.

(21) FIG. 20 is another flow diagram illustrating an example of steps and sub-steps within a

pipeline in accordance with the systems and methods described herein.

(22) FIG. **21** is a flow diagram illustrating an example of steps related to segmentation in accordance with the systems and methods described herein.

(23) FIG. **22** is a diagram related to segmentation in accordance with the systems and methods described herein.

(24) FIG. **23** is a flow diagram illustrating an example related to correspondence in accordance with the systems and methods described herein.

(25) FIGS. **24A-24K** are diagrams related to correspondence in accordance with the systems and methods described herein.

(26) FIG. **25** is another diagram related to correspondence in accordance with the systems and methods described herein.

(27) FIG. **26** is another flow diagram illustrating an example of steps and sub-steps within a pipeline in accordance with the systems and methods described herein.

(28) FIGS. **27A1-27F** are diagrams related to warping in accordance with the systems and methods described herein.

(29) FIG. **28** is another flow diagram illustrating an example of steps and sub-steps within a pipeline in accordance with the systems and methods described herein.

(30) FIGS. **29A-29B** are diagrams related to post-processing in accordance with the systems and methods described herein.

(31) FIG. **30** is a diagram illustrating vision for machine learning virtual try-on in accordance with the systems and methods described herein.

(32) FIG. **31** is a diagram related to limitations of computer vision and/or machine learning in accordance with the systems and methods described herein.

(33) FIG. **32** is a table illustrating data requirements in accordance with the systems and methods described herein.

(34) FIGS. **33A-33E** are diagrams related to example approaches in accordance with the systems and methods described herein.

(35) FIG. **34** is a diagram illustrating a data strategy in accordance with the systems and methods described herein.

(36) FIG. **35** is a diagram illustrating items blocking one or more articles of clothing in accordance with the systems and methods described herein.

(37) FIG. **36** is a diagram illustrating labeling in accordance with the systems and methods described herein.

(38) FIG. **37** is a diagram illustrating an example of data curation within a data pipeline in accordance with the systems and methods described herein.

(39) FIG. **38** is a diagram illustrating posing issues in accordance with the systems and methods described herein.

(40) FIG. **39** is a diagram illustrating mobile styler sizes in accordance with the systems and methods described herein.

(41) FIG. **40** is a diagram illustrating a small mobile styler in accordance with the systems and methods described herein.

(42) FIG. **41** is a diagram illustrating a medium mobile styler in accordance with the systems and methods described herein.

(43) FIG. **42** is a diagram illustrating a large mobile styler in accordance with the systems and methods described herein.

(44) FIG. **43** is a diagram illustrating mobile styler sizes in accordance with the systems and methods described herein.

(45) FIG. **44** is a diagram illustrating a small desktop styler in accordance with the systems and methods described herein.

(46) FIG. **45** is a diagram illustrating a medium desktop styler in accordance with the systems and

methods described herein.

(47) FIG. **46** is a diagram illustrating a large desktop styler in accordance with the systems and methods described herein.

(48) FIG. **47** is a diagram illustrating features of a virtual try-on system in accordance with the systems and methods described herein.

(49) FIG. **48** is a diagram illustrating saving a combination of clothing (e.g., a “look”) in a virtual try-on system in accordance with the systems and methods described herein.

(50) FIGS. **49A-49D** are portions of a flow diagram illustrating an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(51) FIG. **50** is a flow diagram illustrating sections **1.1-1.3** of the example flow diagram of FIGS. **49A-49D** illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein.

(52) FIG. **51** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(53) FIG. **52** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(54) FIG. **53** is a flow diagram illustrating sections **2.1-3.1** of the example flow diagram of FIGS. **49A-49D** illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein.

(55) FIG. **54** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(56) FIG. **55** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(57) FIG. **56** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(58) FIG. **57** is a flow diagram illustrating section **5.0** of the example flow diagram of FIGS. **49A-49D** illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein.

(59) FIG. **58** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(60) FIG. **59** is a flow diagram illustrating sections **6.0-6.1** of the example flow diagram of FIGS. **49A-49D** illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein.

(61) FIG. **60** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein.

(62) FIGS. **61A-61B** form a diagram illustrating styling examples in a virtual try-on system in accordance with the systems and methods described herein.

(63) FIGS. **62A-62AE** forms a diagram illustrating example styling combinations in a virtual try-on system in accordance with the systems and methods described herein.

(64) FIG. **63** is a diagram illustrating example styling combinations in a virtual try-on system in accordance with the systems and methods described herein.

(65) FIG. **64** is a diagram illustrating example styling combinations in a virtual try-on system in accordance with the systems and methods described herein.

(66) FIGS. **65A-65D** are diagrams illustrating example features and functionalities in a virtual try-on system in accordance with the systems and methods described herein.

(67) FIG. **66** is a diagram illustrating example features and functionalities in a virtual try-on system in accordance with the systems and methods described herein.

(68) The figures and the following description describe certain implementations by way of illustration only. One skilled in the art will readily recognize from the following description that alternative implementations of the structures and methods illustrated herein may be employed

without departing from the principles described herein. Reference will now be made in detail to several implementations, examples of which are illustrated in the accompanying figures. It is noted that wherever practicable similar or like reference numbers may be used in the figures to indicate similar or like functionality.

DETAILED DESCRIPTION

(69) The detailed description set forth below in connection with the appended drawings is intended as a description of configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

(70) Some implementations of the systems and methods described herein include a virtual try-on system with enhanced styling capability for clothing and accessories. In some implementations, the present invention relates to a virtual try-on system that allows users to virtually try on multiple clothing items (and accessories) and provides enhanced styling options. The system may employ advanced computer vision and machine learning and augmented reality technologies to create a realistic and immersive virtual try-on experience. By incorporating styling capabilities, users may be able to personalize and customize their virtual outfits to suit their preferences. The system utilizes an intuitive user interface and a vast database of clothing styles and accessories to provide a seamless and enjoyable virtual shopping experience.

(71) In some implementations, the virtual try-on system with enhanced styling capability for clothing described herein enables users to visualize and evaluate the appearance of clothing items without physically wearing them. Some example systems and methods may combine state-of-the-art computer vision algorithms, machine learning algorithms, augmented reality techniques, and a vast clothing style database to create a realistic and interactive virtual try-on experience.

(72) In an example implementation, the virtual try-on process may begin with a user selecting a clothing item from the system's extensive database. The database may include various garments such as tops, bottoms, dresses, jackets, and accessories. Each clothing item may be digitally rendered to accurately represent clothing item's design, color, texture, and fit, and each accessory's design, color, texture, or other attributes of the accessory.

(73) In an example implementation, once the user selects a clothing item, the system may initiate the virtual try-on process. Using computer vision and machine learning algorithms, the system detects and analyzes the user's body shape, size, and proportions based on images or video input. The algorithms map the user's body onto a virtual model, creating a personalized avatar that accurately represents the user's physique.

(74) In some example implementations, the selected clothing item may then be dynamically fitted onto the user's virtual avatar. Advanced simulation algorithms account for fabric draping, stretching, and body movement to provide a realistic representation of how the clothing would appear on the user.

(75) An example implementation may have enhanced styling capabilities. The virtual try-on system may go beyond mere visualization by incorporating enhanced styling capabilities. For example, in some implementations, users may be able to personalize their virtual outfits by selecting different clothing combinations, adjusting garment parameters (e.g., sleeve length, neckline, top button unbuttoned, multiple unbuttoned buttons, or other styling), and experimenting with color variations. In an example implementation, the system may allow users to mix and match various clothing items and/or accessories to create unique ensembles. The unique ensembles may be saved for later reference.

(76) Additionally, the system may provide accessory options such as shoes, bags, jewelry, hats, and/or other accessory options, enabling users to complete their virtual outfits with coordinating

elements. In some examples, users can preview different accessories on their avatar and evaluate the overall styling effect.

(77) In an example implementation, user interface and controls may be used to control the systems and methods described herein. In an example implementation, the virtual try-on system features an intuitive and user-friendly interface that may facilitate easy navigation and customization. Users may interact with the system through a smartphone application, a computer application, a web-based platform, dedicated virtual reality (VR) headsets, or other electronic computing device.

(78) The user interface may provide controls for selecting clothing items, adjusting garment parameters, and exploring styling options. Users can browse through the clothing database, filter items by category, style, color, or brand, and view detailed product descriptions and user reviews.

(79) An example implementation may include integration with one or more e-commerce platforms. To enhance the shopping experience, the virtual try-on system may seamlessly integrate with ecommerce platforms. Users may be able to directly purchase the clothing items they try on or add them to a Wishlist for future reference. The system provides links to online retailers, allowing users to access additional product information and make informed purchase decisions.

(80) In addition to the core virtual try-on and styling capabilities, the system may include one or more supplementary features. These may include social sharing, e.g., users can share their virtual outfits on social media platforms, allowing friends and followers to provide feedback and suggestions. These may also include body measurement assistance. For example, the system may offer tools to help users accurately measure their body dimensions, ensuring better fitting virtual garments. An example may provide a virtual fitting room. In a virtual fitting room, users may be able to create personalized fitting rooms where they can store and organize their favorite clothing items for easy access during subsequent sessions. The virtual try-on system with enhanced styling capability for clothing described herein revolutionizes the way users interact with fashion in the digital.

(81) In some implementations, the present disclosure provides a system and method for virtual try-on of, and subscription access to, digital representations of real-world garments with selected styling using machine learning. The present system enables a user to experience online/virtual try-on of unlimited offered styles/articles of virtual clothing (e.g., representing real-world clothing) within a very short time (e.g., 5-10 seconds) following submission of a photo or image of the user. In an exemplary implementation the present system may utilize a first machine learning pipeline to create style-aware datasets based upon a large set of images of garments obtained from, for example, various open source repositories of images, via paid licenses of images, affiliate company's images, or any other sources of images. These style-aware datasets are then used in a second machine learning pipeline, i.e., a try-on/styling pipeline, to train a set of machine learning models. In one implementation these machine learning models include a segmentation generator, a warping model and a try-on synthesis model. During the training process quantitative and potentially qualitative evaluation of performance of the models may be used to infer or otherwise determine checkpoints to be promoted for each model.

(82) Once the segmentation generator, warping model and try-on synthesis model have been trained, the try-on/styling pipeline may be utilized in conjunction with computer vision and machine learning pre-processing and post-processing modules to facilitate virtual try-on of digital garments, e.g., digital representations of real-world garments, styled in accordance with a fashion standard, e.g., a high-fashion standard. In one implementation the virtual try-on process is initiated when a user application executing on a mobile device sends input information to a server platform (e.g., a cloud-based server or a hosted server) executing the try-on/styling pipeline. This input information may typically include information identifying a ghost image of a selected digital garment, a front facing image of a user desiring to virtually try-on the digital garment, and an indication of a fashion style in which the user desires to “wear” the garment. In accordance with one aspect of the disclosed system the result of the virtual try-on process is an image of the user

wearing the digital garment in the user-selected fashion style.

(83) Once the input information is received at the server platform, a computer vision and machine learning pre-processing operation may be performed pursuant to which various checks are made on the image provided by the user (e.g., only 1 person in the image, only certain poses permitted, etc.). Once these checks are performed various pre-processing operations may also be carried out (e.g., remove background and replace it with a white or other uniform background). These pre-processing operations will also include generating all features required by the segmentation generator, the warping model and the try-on synthesis model. The segmentation generator then utilizes a model (checkpoint) to, based on the selected garment and the initial body segmentation determined during pre-processing, predict the target area of the garment in the segmentation. The trained warping model (checkpoint) then deforms the garment based on the predicted segmentation. The trained try-on synthesis model (checkpoint) then executes, based upon styling customizations conforming to, for example, a high-fashion standard, or some other standard, the predicted segmentation, and the warped garment, transfer of a fashion styled garment to a cloth agnostic representation of the user. This results in a live generated image of the user wearing, in the user-selected style and/or styled look, the fashion styled digital garment in a pose identified from the user's input image in a very short time (e.g., 5 to 10 seconds).

(84) In one implementation post-processing of the generated image is employed to improve the generated machine learning output and to detect if the generated image is of sufficient quality to be returned to the user. For example, the generated image may be evaluated to determine if the image features are not within a distribution of images having styling features corresponding to a particular category of a digital styling taxonomy. The post-processing may also include, for example, performing segmentation and computing body part proportions. Other post-processing may include performing pose detection and identifying any anomalous poses or body configurations (e.g., 3 arms, 2 heads, 3 legs, and so on).

(85) Attention is now directed to FIG. 1, which provides an overview of an exemplary system **100** for virtual try-on of a digital representation of a real-world garment and rental of garments styled in conformity with a fashion standard in accordance with the disclosure. System **100** includes a digital garment virtual try-on and rental platform **110** in communication with a mobile device **116** over a network connection **118**, which may include multiple wireless and/or wired networks. The mobile device **116** executes a user application **120** which, in cooperation with platform **110**, facilitates digital garment try-on and rental in accordance with the disclosure. In certain implementations the user application **120** is also configured to support providing fashion styling recommendations, e.g., high-fashion styling recommendations, to a user of the mobile device **116** and to enable the sharing of images of the user wearing digital garments styled in conformity with the fashion standard.

(86) In one implementation platform **110** may be implemented using “cloud” computing capabilities **112**. As is known, cloud computing may be characterized as a model for facilitating on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. Cloud systems tend to automatically control resource use by utilizing some form of metering capability with respect to, for example, storage, processing, bandwidth, and active user accounts. Various cloud service models are possible, including cloud software as a service (SaaS), cloud platform as a service (PaaS), and cloud infrastructure as a service (IaaS).

(87) In other implementations platform **110** may be implemented by using on-premises servers and other infrastructure rather than by using cloud-based services. Alternatively, hybrid implementations of the attribution computation platform **110** including a combination of on-premises and cloud-based infrastructure are also within the scope of the present disclosure.

(88) As shown in FIG. 1, platform **110** includes a set of machine learning models **124** for enabling virtual try-on of digital representations of real-world garments styled in conformity with a fashion

standard. The machine learning models **124**, which may be trained in the manner described hereinafter, include a segmentation generator **144**, a garment warping model **148**, and a try-on synthesis model **152**. Once trained, the machine-learning models **124** are provided with input images received by a computer vision and machine learning pre-processing module **128** from the user application **120** of the mobile device **116**. As is discussed below, the pre-processing module **128** performs various checks on the received input images and executes various pre-processing operations, including generating features from the input images which are used by the machine learning models **124**.

(89) Based upon the pre-processed input image information generated by the computer vision and machine learning pre-processing module **128**, the trained machine learning models **124** generate an image of a user wearing, in a user-selected style, a user-specified digital garment, e.g., a digital representation of a real-world garment, in a pose identified from the input image provided to the user application **120**. A computer vision and machine learning post-processing module **132** is configured to post-process the image generated by the machine learning models **124** in order to, for example, detect if the generated image is of sufficient quality to be returned to the user of the mobile device **116**. For example, the generated image may be evaluated by the post-processing module **132** in order to determine if the image features are not within a distribution of the images used during the machine learning training process. In certain implementations the post-processing module **132** may be configured to perform other post-processing operations of the type described hereinafter.

(90) As indicated above, given a front facing image of a user of the application **120** and a garment image specified the user via the application **120**, the machine learning models **124** executed by the platform **110** synthesize a new image of the user wearing the target garment. In one implementation the pipeline implemented by the machine learning models **124** does not alter the person's pose, the body shape or the person's identity. Users instead are, in some implementations, encouraged to upload several various solo images of themselves in various different poses, hair styles, hats/headwear etc. via application **120** to allow them the option to garment transfer on to various different looks of themselves. In addition, the virtual try-on experience supports different styling options associated with a fashion standard to apply on the user-selected target garment; for example, the styling options for a target garment in the form of a blouse or shirt could include sleeves pushed up, sleeves cuffed, bodice tucked in, and so on, as contemplated by the fashion standard. As is discussed below, the available styles of the target garment may be defined within a digital styling taxonomy comporting with a fashion standard. To enable the machine learning models **124** to facilitate this style-aware virtual try-on process a set of style-aware garment datasets are generated in the manner described hereinafter. These style-aware datasets may then be utilized in a subsequent machine learning training process.

(91) FIGS. **15A-15O** define an exemplary digital style taxonomy **1500** including a plurality of styling combination categories conforming to a fashion standard, e.g., a high fashion standard. For example, a first styling combination category **1502** includes tops having the following styling segments: sleeves buttoned, half tucked, fully buttoned front portion and collar folded. FIGS. **16A-16J** provide examples of options for the various styling segments included within the styling combination categories of FIGS. **15A-15O**. For example, FIG. **16A** illustrates a Sleeve Length styling segment having potential length values of Wrist, Mid Hand and Fingers. As another example, FIG. **16E** illustrates a Bodice Fit styling segment having potential fit values of Tight, Relaxed and Oversized.

(92) Turning now to FIG. **2**, a process flow diagram is provided which illustrates a machine learning data pipeline **210** and a machine learning try-on/styling pipeline **220** in accordance with the disclosure. The machine learning data pipeline **210** prepares style aware datasets in the manner discussed below. The machine learning try-on/styling pipeline **220** exposes try-on and styling endpoints. In one implementation the learning try-on/styling pipeline **220** includes three primary

layers: a training layer, an evaluation layer and an inference layer. The training layer uses a style aware dataset generated by the machine learning data pipeline **210** in order to train the machine learning models **124**. In one implementation the evaluation layer applies quantitative (structural similarity Index: SSIM, Perceptual similarity, F1-Score, Fréchet Inception Distance: FID) and qualitative evaluations to determine the checkpoint to promote for each of the machine learning models **124**. The inference layer exposes the try-on and styling via API endpoints, it also includes a Try-On/Styling Orchestration logic, Pre/Post processing.

(93) The machine learning data pipeline **210** may be configured to start by capturing a person front facing image, a ghost garment image and PDP (product description page) text from a page of a website. In one implementation this data capture is done using customized web crawlers: The information from each scraped website will generally be scraped or otherwise captured by a dedicated web crawler. This raw, captured data is cleaned, enriched and organized to be ready for large scale machine learning training. In one implementation the WebDataset standard is used as a standard utilizing in build the training dataset. The WebDataset is a PyTorch Dataset (IterableDataset) implementation providing efficient access to datasets stored in POSIX tar archives and uses only sequential/streaming data access. This advantageously brings substantial performance advantage in many compute environments, and is desirable and essential for very large scale training.

(94) Exemplary implementations of the machine learning data pipeline **210** do not use the raw garment and other web image data in the learning process. Rather, the raw data is first selected, manipulated, and then transformed into features to train the machine. In one implementation the process of selecting from the raw, capture data results in exclusion of all corrupted images (e.g., images files that cannot be opened), person image with empty or unusable poses (e.g., poses with persons arms crossed), and all ghost images with an empty or invalid garment. Empty or invalid garment images include but are not limited to placeholder images that are included on websites when an item is no longer available and therefore does not show the intended garment. Also excluded during the selection process are person images with invalid segmentation (e.g., no top, top area height is larger than 70% of overall image height). Manipulation of the web image data may include resizing the captured images to a predefined pixel size, e.g., 1024×768.

(95) In one implementation the captured image data is transformed by generating a body semantic segmentation using QNet and Part Grouping Network (PGN) in accordance with techniques described in, for example, “Instance-level Human Parsing via Part Grouping Network”, Gong et al., ECCV 2018 and implemented on CIHP dataset. The transformation includes generating warped clothing for the try-on synthesis model using the best warping checkpoint. As used herein, the term “checkpoint” refers to a snapshot of a working model during the training process. The cloth-agnostic person image and cloth-agnostic person segmentation are then generated, FIGS. 3A and 3B respectively illustrate exemplary inputs **310** and outputs **320** of the machine learning data pipeline **210**.

(96) In some example implementations, a correspondence between the target and source model may be calculated at this stage of the process. The correspondence between the target and source model is a fundamental step in this method, as it allows the system to warp and align the garment to the target model based on their unique body measurements and features.

(97) Image correspondence in computer vision and machine learning refers to the process of identifying and matching the same features or points in two or more images.

(98) The steps in establishing the correspondence in this method are as follow:

(99) Feature point detection and description. In this step the goal is first to identify distinctive points (key points) in the images that can be easily recognized and matched across different images. Second, to create a descriptor for each keypoint that captures the local appearance or structure around the point. A deep learning method is used to simultaneously both detect the interest points and generate their descriptors. For example, see DeTone, Daniel, Tomasz

Malisiewicz, and Andrew Rabinovich. "Superpoint: Self-supervised interest point detection and description." In Proceedings of the IEEE conference on computer vision and pattern recognition workshops, pp. 224-236. 2018.

(100) SuperPoint uses a single CNN to perform both interest point detection and description tasks. The network is composed of two parts: an encoder, which extracts features from the input image, and two parallel decoders, one for detecting interest points and the other for generating descriptors. The network is trained using a self-supervised approach, eliminating the need for manual annotation. SuperPoint outperforms the state-of-the-art models both in terms of the detection and description performance.

(101) Feature matching and correspondence validation. In this step the goal is first to compare the descriptors from different images and identify the best matches. Second, to remove incorrect matches or outliers using a validation method. A deep learning-based method called SuperGlue may be utilized at this stage. (See Sarlin, Paul-Edouard, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. "Superglue: Learning feature matching with graph neural networks." In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 4938-4947. 2020.) SuperGlue simultaneously performs context aggregation, matching, and filtering in a single end-to-end architecture. SuperGlue is a Graph Neural Network (GNN) that leverages both spatial relationships of the keypoints and their visual appearance through self-(intraimage) and cross-(inter-image) attention. The attention mechanisms used in SuperGlue are inspired by the success of the Transformer architecture. Self-attention boosts the receptive field of local descriptors, while cross-attention enables cross-image communication and is inspired by how humans look back-and-forth when matching images. In addition to SuperGlue, tiling may be used in some implementations. For example, in some implementations, images may be split up to maximize the key points generated. In this procedure (tiling) an image may be split into at least two parts, e.g., two or more parts. In some examples, each part may be the same size. In other examples, each part may be different sizes, as needed. By splitting an image into multiple parts, the number of key points generated may be increased. By reducing the size of the image, e.g., by splitting the image into at least two smaller images, each image may be processed more extensively and thus, the two images generate more meaningful points than the original image.

(102) The correspondence established at this step is highly accurate and in one implementation, returns a minimum of 300 corresponding keypoints. This high number of keypoints ensures that the TPS warping can accurately capture the intricate details of the garment and the target model's body shape. It also helps in accounting for the variations in fabric behavior, draping, and folding patterns, which are essential to create a realistic virtual try-on and styling experience. Some implementations may be a realistic virtual try-on and luxury styling experience. Other implementations may return 200 to 1000 keypoints, 100 to 2000 keypoints, or more keypoints. For example, some implementations may return 100 to 3000 keypoints, 100 to 4000 keypoints, or more.

(103) In some implementations, having a minimum of 300 key points allows the system to distribute the correspondences across the garment and the target model's body more evenly. This distribution ensures that the TPS warping covers a wide range of garment areas and body parts, accounting for different levels of detail and complexity. As a result, the warped garment will more accurately conform to the target model's body shape and exhibit a natural-looking fit. Additionally, this high number of keypoints contributes to the system's robustness against various challenges, ensuring a reliable and high-quality virtual try-on and styling experience standard for a digital representation of a real-world garment.

(104) FIG. 4 provides a flowchart of an exemplary workflow **400** for building the base webdataset and each specific version for the different machine learning models **124**. To build a dataset with a specific style conforming to a fashion standard (e.g., top sleeves pushed up, and tucked-in), images corresponding to the given style are identified from the entire base webdataset. In one

implementation this is a three-stage process. In a first stage, the training, test and evaluation datasets are built. In one implementation this may be done using a taxonomy data tagging/labeling tool **230** (FIG. 2) in the form of, for example, an internal instance of label studio with an already setup labeling project. This may involve utilizing experts to label a large number (e.g., on the order of 1000) of images **232** (FIG. 2) of the target style. In a second stage the classifier **234a** (FIG. 2) is trained and the trained model is evaluated. If, based on the evaluation, it is determined that additional training data is required, then experts may be utilized to label a larger set of images of the target style. In a third “inference” stage, the style classifier is executed against the entire dataset and an appropriate styles classifier checkpoint **234b** is inferred. The output of this third stage is a ready to use fashion-aware dataset **240** which may be used to train the virtual try-on pipeline **220** for the target style.

(105) Turning now to FIG. 5, there is shown an architecture **500** for an exemplary implementation of a multi-class styles classifier **234** which may be trained to recognize garment styles presented to it during a training process. The architecture **500** of the classifier **234** is based on the transfer learning technique for image classification. As used herein, the term transfer learning refers to an approach to machine learning in which a model developed for a task is reused as the starting point for a model on a second task. In one implementation transfer learning is conducted in the present context using Visual Geometry Group (VGG19) with 19 CNN (Convolutional Neural Network) layers, trained with ImageNet dataset.

(106) FIGS. 6 and 7 provide high-level representations of operation and training of a multiclass styles classifier **610** implemented using the classifier architecture **500**. The classifier **610** is configured to assign a style (if applicable) to each image in the training data set **232** (FIG. 2). Specifically, classifier **610** determines, for each image in the data set **232**, a probability **614** of the image depicting a garment of a specific style. To avoid mislabeled data, a probability threshold $P_{threshold}$ is set above which it is determined, during a voting stage **620**, that sufficient confidence exists to assign the matching style variation to the image. This is the situation represented in FIG. 6, in which the probability associated with “style A” was determined to exceed $P_{threshold}$. If the voting **620** step fails (FIG. 7), the image is added to a parking lot **630** accessible to experts capable of reviewing the unclassified image and assigning the correct label. As illustrated in FIG. 7, the image will then be added **640** to the training dataset of the style matching the correct label.

(107) Turning now to FIGS. 8A and 8B, an overview is provided of operation of an exemplary machine-learning try-on/styling pipeline **800** in accordance with the disclosure. The pipeline **800** operates to receive from the user application **120**, a front facing user photo, a selected garment to be digitally worn, and target garment styling options conforming to a fashion standard, e.g., high-fashion standard (stage **804**). Computer vision and machine learning pre-processing may be performed by determining the received photo meets person/pose requirements and an initial body segmentation and other features required by machine learning try-on/styling synthesis are generated (stage **808**). A try-on/styling orchestrator is responsible for routing all the inputs and the generated features to the appropriate machine learning try-on/styling instance (stage **810**).

(108) FIG. 9 illustrates an exemplary try-on/styling orchestrator **900** in accordance with the disclosure. In one implementation the orchestrator **900** has an internal factory that will instantiate the right machine learning models given the style. The orchestrator uses an internal map to determine the right machine learning models to use during the try-on process.

(109) Models trained by machine learning may be used to execute try-on of garments styled in accordance with selected target styling options conforming to a fashion standard (stage **812**). In particular, a model may be used to predict, based on the selected garment and the initial body segmentation, the target area of the selected garment in the segmentation (stage **814**). The selected garment may be warped by using a trained model to deform the garment based on the predicted segmentation (stage **816**). A new image of the user wearing the selected garment may be synthesized by using a trained model to execute a garment transfer process based upon the selected

target styling options conforming to a fashion standard, the predicted segmentation, and the warped garment (stage **820**). In one implementation post-processing is performed to improve the machine learning output and detect, by performing segmentation, pose and anomaly detection if the synthesized image is of sufficient quality to be returned to the user for rendering on the user computing device (stage **828**).

(110) For example, once the corresponding key points between the source and target model is established, a non-affine transformation needs to be computed to warp the garment from the source model onto the target model. Non-affine transformations, as opposed to affine transformations, can handle more complex deformations, allowing for a better adaptation of the garment to the unique contours and features of the target model's body shape.

(111) Thin-plate spline (TPS) warping is one such non-affine transformation technique that is commonly used for this purpose. The TPS transformation is a smooth, invertible mapping function that minimizes a certain bending energy, ensuring that the warped image remains smooth and does not introduce unwanted artifacts or distortions.

(112) To compute the TPS warping, the system first calculates the coefficients of the thin-plate spline function using the established key points correspondence. This involves solving a linear system of equations, which results in a set of coefficients that define the smooth mapping between the key points in the source and target domains.

(113) Next, the system applies the calculated TPS transformation to the entire garment image, not just the key points. This involves interpolating the transformation over the entire image, ensuring that the garment's pixels are smoothly warped onto the target model. This step is crucial for maintaining the visual integrity of the garment while adapting it to the target model's body shape.

(114) An Image inpainting technique is employed to eliminate any visible artifacts or imperfections in the overlaid garment. A diffusion based inpainting method is utilized in this method. This algorithm fills in the imperfections by considering the surrounding texture and color information, resulting in a more visually appealing output. An implementation may use dynamic dilation masks. The dynamic dilation masks may be used to identify artifacts, imperfections, or both artifacts and imperfections that are missing.

(115) The computer vision module adjusts the colors, shading, and textures of the garment to ensure that it matches the lighting and appearance of the target model.

(116) The systems and methods described herein may also perform styling. This process involves lifting a styled garment from the source model to the target model. To ensure the accuracy and realism of this process, semantic tags are utilized on the correspondence garments. These tags indicate landmarks specific to particular styles of clothing and provide positional information about the garment on the source model. By identifying and utilizing these landmarks, semantic tags can guide the lifting and pasting of the garment onto the target model both in the try on and the post processing module of the pipeline. For instance, in a denim jacket, semantic tags can help identify landmarks such as the collar, cuffs, and hemline. Through the incorporation of these landmarks, a virtual try-on system can realistically model the behavior of the jacket on the target model, e.g., a real-world jacket on a real-world target model. This includes simulating details such as where the sleeves would naturally push up when rolled or how the hemline would move when the jacket is half-tucked. By accurately representing these details, the virtual try-on system can create a more immersive and believable try-on experience for the user.

(117) FIG. **10** illustratively represents an exemplary try-on/styling workflow **1000** in accordance with the disclosure. The segmentation generators **144** may be implemented using a generative adversarial network (GAN) with, for example, a convolutional neural network such as a U-Net Generator and multi-scale patch discriminators. See, e.g., "U-Net: Convolutional Networks for Biomedical Image Segmentation," Ronneberger et al., arXiv: 1505.04597v1 [cs.CV] 18 May 2015. As indicated by FIG. **10**, the try-on/styling workflow **1000** utilizes machine learning models to implement segmentation, garment warping, and try-on operations. For example, once the garment

has been warped to fit the target model, the computer vision module overlays the garment onto the target model image. This step involves blending techniques to seamlessly integrate the garment with the target model, ensuring a realistic appearance.

(118) The system performs post-processing steps to refine the visual output further. This module includes several techniques to enhance the final results.

(119) Fine tuning the garment fit. To eliminate any imperfections in the overlaid garment an iterative refinement technique is employed. This method iteratively adjusts the warped garment until the overlaid error falls below a predefined threshold.

(120) The iterative refinement process begins by calculating the error between the overlaid garment and the target model. This error is quantified by subtracting the target model upper body segmentation from the warped garment mask.

(121) During each iteration, the system identifies regions in the overlaid garment that contribute to the error and makes appropriate adjustments. These adjustments involve re-warping the garment locally and modifying the blending weights. The error is then recalculated, and the process repeats until the error falls below the specified threshold or a maximum number of iterations are reached.

(122) FIG. 11 represents a U-Net architecture of a type useful in implementing the segmentation generators 144 utilized in the try-on/styling workflow 1000. A principal objective of the model implemented by segmentation generators 144 is to predict the area of the target garment in the person body segmentation. In inference mode, the inputs to this model are: (i) the cloth agnostic segmentation (FIG. 12C); (ii) the person pose; (iii) the cloth (FIG. 12B); (iv) the cloth mask. During training the scraped images (FIG. 12B) are used to build the cloth agnostic image of the person wearing the target garment, where the ground truth is the person segmentation (FIG. 12D).

(123) Turning now to FIG. 13, the machine learning model for garment warping 148 may be implemented using an architecture comprised of a Geometric Matching Module (GMM) with Thin Plate Spline (TPS). This architecture is described in, for example, in “Toward Characteristic-Preserving Image-based Virtual Try-On Network”, Wang et al., arXiv: 1807.07688v3 [cs.CV] 12 Sep. 2018. The model implemented for garment warping 148 predicts the garment deformations to fill the target body part in the person segmentation. As shown in FIG. 13A, in one implementation the model relies on a grid (e.g., 12×16). This grid is used in order to have a sense of the deformation using the grid key points (e.g., 192 points). During training, the model for garment warping takes the ghost images with no deformation (FIG. 13C), the person pose and segmentation. In this case the ground truth is the garment extract from the person wearing it (FIG. 13B). The garment is extracted using the person semantic segmentation.

(124) In one example implementation, garment extraction may be performed. For example, users select garments from a pre-existing database provided by the retailer. In one example, a deep learning-based segmentation technique, such as in Gong, Ke, Xiaodan Liang, Yicheng Li, Yimin Chen, Ming Yang, and Liang Lin. “Instance-level Human Parsing Via Part Grouping Network” in *Proceedings of the European conference on computer vision (ECCV)*, pp. 770-785. 2018, may be employed to accurately separate the garment from the background and the source model.

(125) In one implementation the machine learning model for try-on synthesis 152 may be implemented using an architecture comprised of a generative adversarial network (GAN) with a pix2pixHD generator and multi-scale patch discriminators. Pix2pixHD is a method for synthesizing high resolution photo-realistic images from semantic label maps using conditional generative adversarial networks (CGANs). See, e.g., “High-Resolution Image Synthesis and Semantic Manipulation with Conditional GANs”, Wang et al., arXiv: 1711.11585v2 [cs.CV] 20; August 2018. In order to improve the overall result the perceptual loss (based a pre-trained VGG16) may be used to ascertain the overall loss function. See, e.g., “Perceptual Losses for Real-Time Style Transfer and Super-Resolution”; Johnson et al., arXiv: 1603.08155v1 [cs.CV] 27; March 2016.

(126) In one implementation the model for try-on synthesis 152 is executed as a final step in completing the try-on/styling process. The model for try-on synthesis 152 operates utilizing the

output of the previous models; that is, a generated segmentation (FIG. 14A) and warped garment (FIG. 14B). The model additionally utilizes the initial cloth-agnostic segmentation and pose (FIG. 14C).

(127) Referring again to FIG. 1, in one implementation the platform **110** further includes a digital garment rental and recommendation engine **136** and an outfit/fashion recommendation engine **140** configured to function with the user application **120** to facilitate rental and recommendation of digital garments, e.g., digital representations of real world garments. The digital garment rental and recommendation engine **136** enables a user to “rent” a digital garment that has been virtually tried on by the user in the manner described above. The rented digital garment may be stored within a digital closet for the user maintained by the user by platform **110** during the duration of the rental term. During this rental term the user may submit, via the user application **120**, different images of the user to the platform **110** and in response receive versions of the different images in which some or all of the user's clothing within the submitted images is replaced by the rented digital garment.

(128) The digital garment rental and recommendation engine **136** also enables a user to invite individuals via a link code or scannable image code to view the user's virtual try-on renderings via a mobile application. For example, with one click the user receiving the shared image can swap their own image with that of the original and see themselves in the same attire. Metadata for the outfit displayed enables any user in the sharing group to link to any item available in the original image.

(129) The recommendation engine **136** can also enable users to create a fashion playlist. For example, from a basket of item(s) the user “shuffles” all permutations into a variety of styling and/or complete outfits which the user RANKS and/or LIKES. Based on rank order the system adds selected item(s) and/or styling choices to the shuffle basket. The resulting permutations, ensembles, individually styled item(s) and/or outfits can be tagged by use-case and user-generated keywords. Users can “save” item(s), individually styled item(s), ensembles and/or outfits and have the ability to share them. User-generated metadata and ranking is used to improve metadata and recommendations and create rank order “most popular” individually styled item(s) ensembles and/or outfits that can be further up-voted by the user community. Subtle semantic metadata can be gleaned from user keywords including mood, use case, style, demographic, psychographic, and other explicit and implicit factors.

(130) The outfit/fashion recommendation engine **140** enables outfits to be selected and recommended based on mood. The metadata schema categorizes clothing by use cases taxonomy and emotional vector valence. Via user input, ensembles are assembled based on the use case taxonomy, mood and mood intensity. Users rank order recommendation results to improve precision of subsequent recommendations. Interactive environments are in need for believable characters. Their personality and emotions should affect the behavior in order to increase the emotional immersion of the player. In one proposed model, virtual characters (or agents) are equipped with an individual personality and dynamic emotions. Both models interact within the selection of appropriate actions, and are thus central for the exhibited behavior of autonomous agents. The proposed model is application-independent and allows to render a rich set of behaviors for virtual characters in learning environments. See, e.g., “Emotional Agents for Interactive Environments”; Masuch et al.; 4th Int. Conf. on Creating, Connecting and Collaborating through Computing (C5 2006), Jan. 26-27, 2006, Berkeley, California.

(131) FIG. 17 is a flow diagram illustrating an example of steps **1700** within a pipeline in accordance with the systems and methods described herein. Step **1** within the pipeline is a photo shoot (**1702**). Step **2** within the pipeline is asset generation (**1704**). Step three within the pipeline is multi-stage warping (**1706**). Step **4** within the pipeline is post processing (**1708**). Each of the steps within the example pipeline of FIG. 17 will be discussed in more detail below with respect to FIGS. 18-29B.

(132) FIG. 18 is a flow diagram illustrating an example of steps and sub-steps within a pipeline in

accordance with the systems and methods described herein. More specifically, FIG. **18** illustrates sub-steps within the photo shoot **1702**. The photo shoot **1702** may include capturing initial undergarment shots and/or reference (**1802**). In an example implementation, capturing initial undergarment shots and/or reference (**1802**) may be for all models. A reference shot may refer to a photograph taken as a visual reference or guide for a specific purpose. The reference shot may serve as a benchmark or template that the systems and methods described herein may refer back to as a reference. The reference shot may be a shot of a model (or models) in undergarments.

(133) The photo shoot **1702** may include taking one or more correspondence shots. The correspondence shots may align to the reference shot. For example, in some implementations, the correspondence shots may align to the reference shot for all models **1804**. The correspondence shot may be a type of photograph that captures a subject from different angles, distances, or perspectives. The correspondence shot may involve taking multiple shots of the same subject, and may include varying the composition, framing, or camera settings between each shot. The correspondence shots may provide a range of options for selecting the most visually appealing or compelling image from the series. By experimenting with different angles, perspectives, and settings, photographers may explore creative possibilities and find the best representation of the subject.

(134) The photo shoot **1702** may include taking one or more cloth shots aligning to the reference shot source model only (**1806**). A cloth shot may refer to a photograph that focuses primarily on showcasing a particular item of clothing or fabric. The cloth shot may be a close-up or detailed shot that may highlight the texture, pattern, design, or other unique features of the cloth. When capturing a cloth shot, the photographer may aim to capture the essence and visual appeal of the fabric. This may involve zooming in on the fabric to reveal intricate details, emphasizing the texture by using lighting techniques, or capturing the fabric in a way that highlights one or more of the fabric's drape, flow, or movement. Cloth shots are commonly used in various fields such as fashion photography, textile industry, product catalogs, and advertising. The cloth shot may serve the purpose of showcasing the quality, craftsmanship, and aesthetic aspects of the fabric, enabling viewers to get a closer look at its characteristics. By focusing on the cloth itself, a cloth shot may communicate information about the material, color, pattern, and overall style. The cloth shot may allow viewers to assess the fabric's visual appeal, texture, and suitability for different purposes or fashion applications. The cloth shot in a photo shoot may refer to a specialized photograph that highlights the unique qualities of a fabric, allowing viewers to appreciate the fabric's details, texture, and overall aesthetic appeal.

(135) Each of the shots, e.g., the reference shots, the correspondence shots, the cloth shots, or some combination of the reference shots, the correspondence shots, and the cloth shots may be stored in database **1808**. The database **1808** may be stored on various types of hardware depending on the specific requirements and scale of the database system. Some common types of hardware used to store databases include servers, Storage Area Networks (SAN), Solid-State Drives (SSDs), Redundant Array of Independent Disks (RAID), cloud storage, or some combination of servers, SAN, SSDs, RAID, and cloud storage. While database **1808** may be stored on hardware, database **1808** may be managed and accessed through a database management system (DBMS) software, which provides the interface for interacting with the data stored on the hardware infrastructure.

(136) FIGS. **19A-19G** are diagrams illustrating example photos of a photo shoot in accordance with the systems and methods described herein. FIGS. **19A-19B** illustrate an initial reference shot and subsequent shots. In an example implementation color coding may be used with respect to key points in the pose. For example, red may indicate misaligned key points and green may indicate aligned key points. As illustrated in FIG. **19**, dots may be used to illustrate misalignment and alignment. For example, in color photos, a red dot may indicate misalignment and a green dot may indicate alignment. Additionally, in the illustrated example the silhouette of the initial reference shot may be overlaid on the preview. The silhouette of the initial reference shot may provide better

overall alignment guidance.

(137) FIG. **19C** illustrates various photographs (shots) of a source model in a photo shoot, including a reference shot, a correspondence shot, and a cloth shot. The reference shot, the correspondence shot, and the cloth shot are defined in the discussion of FIG. **18** above. FIG. **19D** illustrates various photographs (shots) of a target model in a photo shoot, including a reference shot and a correspondence shot. FIG. **19E** illustrates misalignment issues between the models. For example, a well aligned shot **1925** and a misaligned shot **1827** are illustrated. FIG. **19F** also illustrates misalignment issues between the models and includes a cloth overlay **1929**. In some example implementations alignment may be performed manually. For example, some example implementations may use manual alignment through the data pipeline. In such an example, a person operating the system may manually align the shots as illustrated in FIG. **19G**.

(138) In a virtual try-on system, manual alignment of a shot may refer to the process of manually adjusting or aligning a virtual garment or accessory, or manually adjusting or aligning a virtual representation of a person with, e.g., a reference shot of a model. In an example, a corresponding user's body in a captured or uploaded image or video. The manual alignment May involve carefully positioning and matching the virtual item (or model) with the user's or model's body landmarks or reference points to create a realistic and accurate virtual representation.

(139) Steps may include User Image Acquisition, Landmark Detection, Virtual Garment Placement, Fine-Tuning and Refinement, Real-Time Visualization. By performing manual alignment of a shot in a virtual try-on system, the aim is to create a seamless integration of the virtual garment or accessory with the user's image or video, providing an accurate representation of how the clothing item would look on the user.

(140) In other example implementations A computing system may compare two or more shots and perform an automated alignment. Automated alignment of two shots in a virtual try-on system may refer to the process of automatically aligning a virtual garment or accessory with the user's body in two separate images or videos. Automated alignment may involve utilizing computer vision and image processing techniques to analyze and align the virtual item with the user's body in a consistent and accurate manner (or aligning a pose of a second model, e.g., the user with a first model, e.g., a reference model). Machine learning may also be used to analyze and align the virtual item with the user's body in a consistent and accurate manner.

(141) Automated alignment may include Landmark Detection, where the system may use computer vision and machine learning algorithms to automatically detect and identify key landmarks or reference points on the user's body in both images and/or videos. These landmarks may act as anchor points for alignment. The system may perform Feature Matching where the system analyzes the features of the virtual garment and the user's body in both images or videos and identifies common features or patterns that can be matched between the two sets of data. The system may also perform Transformation Estimation based on the matched features and detected landmarks, the system estimates the necessary transformations, such as translation, rotation, and scaling, to align the virtual garment with the user's body. Some example implementations may perform Alignment Adjustment. The estimated transformations may be applied to the virtual garment to align it with the user's body in the second image or video. This adjustment ensures that the virtual garment maintains consistent positioning and proportions across the two shots. An example implementation may perform Real-Time Visualization. The system provides real-time visualization of the aligned virtual garment, allowing the user or operator to see how it fits on the user's body in the second image or video. This visualization aids in assessing the alignment quality and making any necessary refinements. Automated alignment in a virtual try-on system may reduce the need for manual intervention and streamlines the process of aligning virtual garments with the user's body. By leveraging computer vision and machine learning techniques, it enables efficient and accurate alignment, providing users with a realistic and personalized virtual try-on experience.

(142) FIG. **20** is another flow diagram illustrating an example of steps and sub-steps **1800** within a

pipeline in accordance with the systems and methods described herein. The sub-steps **1800** include segmentation (**1802**), an Self Correction for Human Parsing (SCHP) model may provide crude body/garment segmentation (**1804**), a determination may be made that an Intersection over Union (IOU) between SCHP segmentation and Segment Anything Model (SAM) masks is greater than a threshold (**1806**). When the IOU between SCHP segmentation and the SAM masks are greater than a threshold (**1806**), the SAM generated masks and masked images may be stored in database (**1808**). SAM generates all possible object masks. The system may perform correspondence generation (**1812**), correspondence may be determined between source and all other models (**1814**). The system may also perform custom filtering on correspondence points (**1816**). Correspondence key points may be in a NumPy array.

(143) IOU may be used as a metric in computer vision, including virtual try-on systems. IOU may be used to measure the overlap between two bounding boxes or regions of interest. For example, IOU may be used to evaluate the accuracy of the virtual garment's alignment with the user's body. IOU may quantify how well the virtual garment aligns with the corresponding region of the user's body in terms of spatial overlap.

(144) IOU may be calculated using a Bounding Box Definition, an Intersection Calculation, a Union Calculation, or a IOU Calculation. An example formula for IOU is: $IOU = \frac{\text{Intersection Area}}{\text{Union Area}}$.

(145) The IOU score may range from 0 to 1, with a score of 1 indicating a perfect alignment or complete overlap between the virtual garment and the user's body. A higher IOU score signifies a better alignment and a closer fit of the virtual garment to the user's body.

(146) Self-Correction for Human Parsing (SCHP) is a body-part and garment segmentation model which provides semantic labels and segmentation masks for person images. (See Li, Peike, Yunqiu Xu, Yunchao Wei, and Yi Yang. "Self-correction for human parsing." IEEE Transactions on Pattern Analysis and Machine Intelligence 44, no. 6 (2020): 3260-3271.) However, its masks are crude and often inaccurate.

(147) Segment Anything Model (SAM) is another segmentation model that only provides segmentation masks for objects, including persons. (See Kirillov, Alexander, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao et al. "Segment anything." arXiv preprint arXiv: 2304.02643 (2023).) SAM provides accurate object masks but it does not provide semantic labels for object masks. By incorporating SCHP guidance with SAM, this improves the localization of objects while providing semantic labels to them.

(148) FIG. **21** is a flow diagram illustrating an example of steps related to segmentation **2100** in accordance with the systems and methods described herein. Segmentation may begin at block (**2102**). The SCHP model may provide crude body and/or garment segmentation (**2104**). A determination may be made that the IOU between the SCHP segmentation and SAM masks is greater than a threshold (**2106**). SAM may generate all possible object masks (**2108**). The SAM generated masks and mask images may be stored in a database **2110**.

(149) FIG. **22** is a diagram related to segmentation in accordance with the systems and methods described herein. FIG. **22** further illustrates segmentation with SCHP-guided SAM. Segmentation may be used to identify which pixels in an image correspond to specific predefined labels.

(150) FIG. **23** is a flow diagram illustrating an example **2300** related to correspondence in accordance with the systems and methods described herein. Correspondence generation may begin at block (**2302**). In virtual try-on, correspondence refers to the process of establishing a meaningful relationship or association between the virtual garment and the user's body or image/video. Correspondence involves identifying and aligning specific points or regions of the garment with the corresponding points or regions on the user's body. The correspondence between source and all other models may be determined (**2304**). Custom filtering on correspondence points may be performed (**2306**). Custom filtering in correspondence for virtual try-on may refer to a technique or process that allows for the customization and refinement of the correspondence between the virtual

garment and the user's body. The filtering may involve applying specific filters or adjustments to improve the accuracy and quality of the alignment. The correspondence keypoints may be stored in a database, e.g., using a numpy array (2308). A Numpy array is a fundamental data structure in the NumPy (Numerical Python) library, which is a popular package for numerical computations in Python. A NumPy array, also known as ndarray, is a multi-dimensional container that stores elements of the same data type in a contiguous block of memory.

(151) FIGS. 24A-24K are diagrams related to correspondence in accordance with the systems and methods described herein. FIG. 24A illustrates transferring correspondence points on an undergarment. FIG. 24B illustrates transferring correspondence points using off-the-self complex patterns. The example of version 1 (FIG. 24B) used a top only. FIG. 24C illustrates transferring correspondence points using a custom top that may have a pattern, and possibly a complex pattern in some examples. The example of version 2 (FIG. 24C) used a top only. FIG. 24D illustrates version 3 for transferring correspondence points using custom garments, e.g., a full bodysuit, in accordance with the systems and methods described herein. The example of version 3 may use a better off-the-shelf full bodysuit. FIG. 24E illustrates transferring correspondence points in version 4, using a custom garments that includes a custom top and a custom bottom in accordance with the systems and methods described herein. FIG. 24F illustrates transferring correspondence points using April Tag Focus in accordance with the systems and methods described herein. FIG. 24G illustrates transferring correspondence points using custom printed and/or stitched garments, e.g., April Tags and/or a triangle pattern in accordance with the systems and methods described herein. FIG. 24H illustrates transferring correspondence points in accordance with the systems and methods described herein. The example indicates that not all points on a garment are created equal. Some points are out of view. Some points are stretched apart, some points are closer together, e.g., based on body shape. FIG. 24I illustrates transferring correspondence points in accordance with the systems and methods described herein. FIG. 24J illustrates a seventh version for determining correspondence on complex patterns. Some examples may used "SuperGlue." Superglue is a specific algorithm or method used for establishing dense pixel-level correspondence between the virtual garment and the user's body or image/video.

(152) FIG. 24K illustrates a custom garment for virtual try-on of fashion garments. In an example implementation, the custom garment may include a fabric or material suitable for garment production. The fabric may form a form-fitting garment. The fabric may include a set of patterns integrated into the fabric. The patterns may be designed for body mapping. For example, the set of patterns may be configured to allow for a mapping system to digitally map a user's body measurements to the set of patterns such that a virtual try-on application may overlay another garment on the user's body based on the mapping using the custom garment.

(153) In an example implementation, the custom garment may be a full body, form-fitting suit, covering a user's body leaving only the hands and head exposed. In an example implementation, the set of patterns integrated into the fabric further comprises one or more of a proprietary mix of colors, a proprietary mix of patterns, and AprilTags. In an example implementation, the set of patterns integrated into the fabric are configured to maximize correspondence points between two different people when both parties wear such a custom garment. In an example implementation, the set of patterns integrated into the fabric are configured to maximize correspondence points between two different people when one party wears the custom garment, and a corresponding custom garment is applied to a second party digitally. In an example implementation, the set of patterns integrated into the fabric further comprises one or more of a proprietary mix of colors, a proprietary mix of patterns, and AprilTags, wherein the set of patterns integrated into the fabric are configured to maximize correspondence points between two different people when a first party and a second party both wear the custom garment digitally.

(154) FIG. 25 is another diagram related to correspondence in accordance with the systems and methods described herein. The figure illustrates a number of different shapes, lines, colors, and

areas. The example uses a full bodysuit. It will be understood that these shapes, lines, colors, and areas can be used on other garments, e.g., tops, bottoms, or any other garment. Generally covering a greater portion of a user's body may be preferable. Additionally, typically covering the area that is generally covered by clothing may also be preferable.

(155) FIG. **26** is another flow diagram illustrating an example of steps and sub-steps within a pipeline in accordance with the systems and methods described herein. More specifically, FIG. **26** illustrates sub-steps to Step **3**, multi dash stage warping. In the illustrated example coarse warping using the stored correspondence for the target model may be performed (**2602**). Fine warping based on differences between body segmentation and warped garment mask may be performed (**2604**). Fusion of warped cloth and target model image may be performed (**2606**). The images may be stored in database **2608**.

(156) FIGS. **27A1-27F** are diagrams related to warping in accordance with the systems and methods described herein. In the illustrated example of FIGS. **27A1-27A2**, the warping method of perspective may be compared to thin plate splines (TPS) between the perspective warp and the TPS warp misalignments may be commonly seen during the perspective warp. FIG. **27B** illustrates the puppet warp and the TPS warp. FIG. **27C** illustrates a coarse warp. FIG. **27D** illustrates a fine warp (left leg). FIG. **27E** illustrates a fine warp (right leg). FIG. **27E** illustrates a fine warp (before and after).

(157) FIG. **28** is another flow diagram illustrating an example of steps and sub-steps **2800** within a pipeline in accordance with the systems and methods described herein. The post-processing (**1708**) may include building a dynamic dilation mask (**2802**), feeding a try-on image and dilation mask to an inpainting model (**2804**), and storing one or more of the post processed images in the database (**2806**). The dynamic dilation masks is used to identify artifacts, imperfections, or both artifacts and imperfections that are missing. Inpainting is discussed below.

(158) FIGS. **29A-29B** are diagrams related to post-processing in accordance with the systems and methods described herein. Inpainting, is a technique used in computer vision and image processing to fill in the missing or occluded regions of a person's appearance when trying on virtual garments or accessories. Inpainting aims to create a visually realistic representation of a person wearing the virtual items, even when certain parts of the body are hidden or obstructed. An example inpainting model used is Stable Diffusion; see Rombach, Robin, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. "High-resolution image synthesis with latent diffusion models." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10684-10695. 2022. FIG. **29A** illustrates inpainting, including an example dynamic dilation mask. FIG. **29B** illustrates the inpainting in the context of other steps, e.g., image processing, asset generation, correspondence and warping, and inpainting.

(159) FIG. **30** is a diagram illustrating vision for machine learning virtual try-on in accordance with the systems and methods described herein. In the illustrated example, a user uploads a photo and selects a look styling and garments (**3002**). The user uploaded photo may then be processed using virtual try-on technology described herein (**3004**) to generate a picture of a person in the uploaded photograph in the selected look, styling and garments (**3006**). An example system may support minimal upload restrictions. For example, some examples may include one or more of dynamic poses, any hairstyle, and/or any skin tone. All garment types and accessories may be supported, and may include layering and styling. Through business to business connections many different garments may be provided.

(160) FIG. **31** is a diagram related to limitations of computer vision and/or machine learning in accordance with the systems and methods described herein. While computer vision and/or machine learning has made significant advancements in virtual try-on applications, there are still some limitations and challenges that researchers and developers face. In some examples, computer vision and/or machine learning may not be able to generate content such as draping wrinkles and/or shadows.

(161) FIG. 32 is a table illustrating data requirements in accordance with the systems and methods described herein. The figure illustrates data requirements of Computer Vision (CV) versus Machine Learning (ML). The data quality for CV may be near pixel perfect. The data quality for ML may be higher and better. The ML data quality may be sufficient volume of quality can overcome imperfections. The data characteristic of CV may be that every piece contains exactly everything required. The data characteristic for ML may be that pieces contain disparate information that needs to be aggregated. The data volume for CV may be linear to range of products, poses, and/or body shapes required. The data volume for ML may be ideally the same as for CV, but may depend on the desired generalization (for scale).

(162) FIGS. 33A-33E are diagrams related to example approaches in accordance with the systems and methods described herein. The figures illustrate the photoshoot, asset generation, and multi-stage warping. The photoshoot may include a source and a target. Asset generation may perform correspondence mapping and source garment extraction. The multi-stage warping may be performed from the output of the correspondence mapping and the source garment extraction. FIG. 33B also includes a user upload and model selection. FIG. 33C also includes a user upload and model selection as well as skin generation, which may generate an input for the warping block. The asset generation of FIGS. 33D-33E may include a multiple image sequence as well as a body mesh and source garment renderer.

(163) FIG. 34 is a diagram illustrating a data strategy in accordance with the systems and methods described herein. The data strategy for supporting ML may include prioritization, acquisition, quality assurance, catalog and version control, and analytics and reporting.

(164) FIG. 35 is a diagram illustrating items blocking one or more articles of clothing in accordance with the systems and methods described herein. In FIG. 35, a jacket may block portions of the pants and shirt, as illustrated. Similarly, a jacket may block parts of the arms and parts of the skirt. For example, the arms may be blocked in a non-standard way, e.g., a middle section of the arms may be visible while the top of the arms and the lower portion of the arms may be blocked.

(165) FIG. 36 is a diagram illustrating labeling in accordance with the systems and methods described herein. The figure illustrates initial labeling and label verification. In some cases, a system may first use AWS's Mechanical Turk service to get an initial labeling on our data. Then, the system may, for further refinement, pass it off to a second pass of more highly trained labelers.

(166) FIG. 37 is a diagram illustrating an example of data curation within a data pipeline in accordance with the systems and methods described herein. In an example implementation, a system may perform asset generation. The output of the asset generation may be prefiltered. The prefilters may include eligibility filters. Filtering may include critical rejections and conditional rejections. With critical rejections data and or assets may be deleted. With conditional rejections images may be stored in a quarantine bucket. These may include PDP tag information, pre filtering information, and assets not quality control (QC) stored in dataset builder. For example, the system may have some assets in its dataset that have yet to be verified by a quality control (QC) person and/or algorithm. Those assets may be put into a quarantine bucket to ensure that those asset's status as not validated is well-known and documented. Assets that pass all pre-filtering may then be stored in a "live" bucket, e.g., images and/or other assets may be stored in a live bucket, including PDP tag information, pre-filtering information, assets not QC stored in the dataset builder. Assets, e.g., segmentations, masks, pose annotation information, that were automatically generated using computer vision techniques or machine learning models are initially marked as "Not QC" until it has been processed through the quality control process. This QC process may be similar to FIG. 36, where an initial assessment of the quality is provided but is subsequently verified.

(167) FIG. 38 is a diagram illustrating posing issues in accordance with the systems and methods described herein. Pose issues may occur in training data. For example, virtual try-on systems may rely on pre-captured training data to model the interaction between garments and the human body. The training data may cover a limited range of poses and postures. Accordingly, issues may occur

when a user does not pose using the limited range of poses and postures used by models for real world garments on the model that have been turned into virtual garments for virtual try-on by the user.

(168) FIG. **39** is a diagram illustrating mobile styler sizes in accordance with the systems and methods described herein. You run on a the mobile styler may be run on, for example a mobile telephone handset, or other mobile electronic device. The mobile styler may come in various sizes. The illustrated examples come in small (FIG. **40**), medium (FIG. **41**), and large (FIG. **42**). It will be understood however that the mobile styler can come in many other sizes.

(169) FIG. **40** is a diagram illustrating a small mobile styler in accordance with the systems and methods described herein. In the illustrated example features of the small mobile styler may include seeing items tried on in a condensed format and maximizing to show medium styler.

(170) FIG. **41** is a diagram illustrating a medium mobile styler in accordance with the systems and methods described herein. In the illustrated example, features of the medium mobile styler may include styling variations, adding items to a bag, adding outfits to saved looks, minimizing to the small styler and maximizing to the large styler.

(171) FIG. **42** is a diagram illustrating a large mobile styler in accordance with the systems and methods described herein. In the illustrated example the features of the large mobile styler include styling an outfit with recommendations, filtering recent products, adding to the cart and favorite looks.

(172) FIG. **43** is a diagram illustrating desktop styler sizes in accordance with the systems and methods described herein. The desktop styler may run on a laptop computer, desktop computer, or other computing platforms. The desktop styler may come in many different sizes. The illustrated examples include a small desktop styler (FIG. **44**), a medium desktop styler (FIG. **45**), and a large desktop styler (FIG. **46**).

(173) FIG. **44** is a diagram illustrating a small desktop styler in accordance with the systems and methods described herein. The small desktop styler includes features such as seeing items tried on in a condensed format and maximizing to medium styler.

(174) FIG. **45** is a diagram illustrating a medium desktop styler in accordance with the systems and methods described herein. The medium desktop styler may include features such as styling variations, adding items to a bag, adding outfits to saved looks, minimizing to the smaller styler, and maximizing to the larger styler.

(175) FIG. **46** is a diagram illustrating a large desktop styler in accordance with the systems and methods described herein. The large desktop styler may include features such as styling variations, adding items to a bag, adding outfits to saved looks, and minimizing to the medium styler as well as filtering recommended products.

(176) FIG. **47** is a diagram illustrating features of a virtual try-on system in accordance with the systems and methods described herein. Features of the systems and methods described herein may include “complete the look” and “style the look.” “Complete the look” may automatically place the recommended outfit on the selected model or image and default to styling recommendations. “Style the look” may style variations offered from a team of expert stylists for each garment or outfit or a style as determined by the user, by another stylist, by an AI stylist.

(177) FIG. **48** is a diagram illustrating saving a combination of clothing (e.g., a “look”) in a virtual try-on system in accordance with the systems and methods described herein. Combinations of clothing also referred to as a look, may be saved so that the look may be referred to later. In the illustrated example the star in the upper right hand corner of the screen just below the button to close the window may be used to add a combination of clothing (a look) to the saved looks. When a look is not saved the star may not be filled in. When a look has been added to the saved looks, the star may be filled in. In some examples the star may be color-coded. For example, the star may be red when a look is saved. It will be understood, of course, that any other color can be used as long as it is different from the background color immediately surrounding the star.

(178) FIGS. **49A-49D** are portions of a flow diagram illustrating an example user flow in a virtual try-on system in accordance with the systems and methods described herein. The user flow will be discussed in more detail below.

(179) FIG. **50** is a flow diagram illustrating sections **1.1-1.3** of the example flow diagram of FIGS. **49A-49D** illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein. The figure illustrates image model selection (**1.0**). For virtual try-on and luxury styling a user may upload a photo (**1.1**). The image may be used for model selection. As another alternative, a user image may be captured from social media (**1.2**). The image captured from social media may be used for model selection. In another example, a model may be selected directly (**1.3**).

(180) FIG. **51** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein. In some example implementations features include Bing background from uploaded image. For example, the systems and methods described herein may automatically remove any and all background from uploaded images of applying the item selected for try-on to the model. As discussed above, in some examples an image may be used from social media. The image from social media may have its background removed, e.g., automatically.

(181) FIG. **52** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein. The figure illustrates model selection. In some examples the systems and methods described herein may fit a model for try-on. The illustrated example a model may be selected closest to the size and shape of the user for try-on and styling. Models may be available in a large variety of sizes such that a wide variety of users may be able to select a model having a similar size and shape to themselves.

(182) As illustrated in FIG. **52**, and discussed above, a user (or some other person) may select a model that is sized and shaped like them. The selected model may then be the “target,” see FIGS. **33A-33E**, and the corresponding discussion, for example. The user (or some other person) may further select a picture of themselves, i.e., the user. The selected photo of the user will then become the new “target.” The user (or some other person) does not directly select what becomes the “source,” see FIGS. **33A-33E**, and the corresponding discussion, for example.

(183) In another implementation, one or more of the photos may be selected by the systems and methods described herein. For example, the system may take a photo of the user and then compare attributes of that user, e.g., based on the photo of that user to select a model that has the same or similar attributes, e.g., is about the same size as the user. For example, estimated measurements of the model(s) and the user, based on estimates from the various photos may be compared.

Furthermore, one or more of the model(s) and/or the user may wear the custom garment described herein for one or more of the photos used to select the model. For example, in some implementations a number of models may wear custom garments and the user may wear a custom garment. The custom garments may provide better estimates regarding the size and shape of the various models and the user. Accordingly, a better model selection may be obtained.

(184) FIG. **53** is a flow diagram illustrating sections **2.1-3.1** of the example flow diagram of FIGS. **49A-49D** illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein. Styling (**2.0**) may include completing the look (**2.1**) where a product may be recommended (**2.3**), e.g., a bag may be added (**2.2**). The system is and methods described herein may include styling the look (**2.4**) and storing the look in a database (**5.0**). Styling the look (**2.4**) may include a favorite look (**3.1**) and adding items tried into a “bag” (**3.2**), e.g., for purchase in an electronic purchasing system. Styling the look (**2.4**) May also lead to removing an item or a layer (**3.3**), removing a favorite item or going to a bag or cart (e.g., an online purchasing storage location). As illustrated in the figure, a user may cycle through the recommended products.

(185) FIG. **54** is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein. As illustrated in the figure, the systems

and methods described herein may include features such as complete the look which may be a recommendation tool that may be used by the user to automatically place the recommended outfit on the selected model or image and default to a predetermined styling recommendation. The systems and methods described herein may also include a recommended product feed. For example these systems and methods may analyze customer data and point to provide the most accurate recommendations that the customer can select for try-on. Items that are tried on may easily be added to a cart.

(186) FIG. 55 is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein. As illustrated in the figure, styled looks may be available to the user. The styled look may be saved. Such an outfit or outfits may be kept in a predetermined area, e.g., a saved looks section of the application, accessible by the user. These styled looks may provide products tried on in an outfit and have the outfit available for purchase. The outfit may also be displayed in an image that is styled for easy reference. A gallery of all a user's tried on looks may be provided. The systems and methods described herein may provide styling variations. The styling variations may be from an expert team of stylists. The styling variations may be provided for each garment or outfit.

(187) FIG. 56 is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein. As illustrated in the figure, a user may try-on items using a styler. The styler may allow the user to see how they will look in any available garment. The systems and methods described herein may be used by the user throughout the shopping experience. In the illustrated example, a user may use the small styler for simple try-ons of items, expand to the medium styler for styling and adding to the cart, and favoriting a look, or open up the large styler for all of the options.

(188) FIG. 57 is a flow diagram illustrating section 5.0 of the example flow diagram of FIGS. 49A-49D illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein. The figure generally illustrates that various points in the methods described herein may lead to saving of data. For example, styling the look (2.4) may lead to saving the look in the customized client dashboard data database (5.0). Cycling through recommended products may also lead to saving in the customized client dashboard data database (5.0). Trying on items from the PLP may also lead to saving data in the customized client dashboard data database (5.0).

(189) FIG. 58 is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein. The figure illustrates A customized client dashboard for data, allowing access to data by the user. The systems and methods described herein may enable new and unique levels of customer insights powered by usage of the styler. Natural language AI powered queries may be possible in some implementations. Direct access to opt in customers during live shopping sessions may be available in some environments. The systems and methods described here and make provide unique shopping insights related to all the fashion experience including outfit preferences category preferences, brand preferences, material preferences and any other user preferences. In some examples, and ability to assess impact of styler usage to AOV return reductions and ESG impact may be provided.

(190) FIG. 59 is a flow diagram illustrating sections 6.0-6.1 of the example flow diagram of FIGS. 49A-49D illustrating a user flow in a virtual try-on system in accordance with the systems and methods described herein. As illustrated in the figure some example implementations may include customized e-mail marketing (6.1). Customized e-mail marketing may provide continued shopping from customized e-mail marketing emails, add advertised looks to saved looks for later, edit and or restyle advertised outfits or pieces, and add advertised outfits to a cart and checkout. The e-mail marketing may also be linked to check out, save looks, and to continue shopping.

(191) FIG. 60 is a diagram illustrating aspects of an example user flow in a virtual try-on system in accordance with the systems and methods described herein. In customized e-mail marketing, as illustrated in the figure, a customer may log in or an account may be required. The system may

utilize data and photos acquired during usage of the systems and methods described herein to personalize communication with customers that have opted-in to the e-mail marketing, for example, a reminder of saved looks may be e-mailed to the user using that user's personal photos to present recommendations outfits and styling options.

(192) FIGS. **61A-61B** form a diagram illustrating styling examples in a virtual try-on system in accordance with the systems and methods described herein. The figure illustrates various example stylings. For example, multiple sleeve stylings, tuck stylings, and button stylings are provided. Additionally different layering stylings are provided as well as color stylings and pant leg stylings. Sleeve stylings may include down, pushed up, rolled, or cuffed. Tuck stylings may include untucked, tucked, front tucked, or half tucked, for example. Button stylings may include top unbuttoned, fully unbuttoned, top and bottom unbuttoned, deep unbuttoned, fully buttoned, or bottom unbuttoned comma for example. Layering can include the use of items such as jackets or shirts for draping and other layered looks. Collar styling can include folded, popped, bow tied, not tied, or untied, for example. Pant leg styling can include down, cuffed, or rolled, for example.

(193) FIGS. **62A-62AE** form a diagram illustrating example styling combinations in a virtual try-on system in accordance with the systems and methods described herein. The figure includes different examples of combinations, including sleeves, tucks, buttons, and collars, as well as a column for describing the combinations.

(194) FIG. **63** is a diagram illustrating example styling combinations in a virtual try-on system in accordance with the systems and methods described herein. The figure includes a table for type, styling, and combinations.

(195) FIG. **64** is another diagram illustrating example styling combinations in a virtual try-on system in accordance with the systems and methods described herein. The figure also includes a table for type, styling, and combinations.

(196) Some implementations relate to a virtual styling system for clothing that enables users to virtually style and customize their outfits. The system incorporates advanced technologies, including computer vision, augmented reality, and artificial intelligence, to provide an immersive and interactive styling experience. Users can select clothing items, experiment with different combinations, adjust parameters, and visualize the styling effects in a realistic virtual environment. The system also offers personalized recommendations and suggestions to assist users in creating unique and fashionable ensembles.

(197) Some implementations of the virtual styling system for clothing described herein revolutionizes the way users engage with fashion by providing a dynamic and interactive platform for virtual styling. The system utilizes cutting-edge technologies to simulate real-world clothing interactions, allowing users to explore various styles and personalize their outfits. Through an intuitive user interface and intelligent algorithms, users can experiment, receive recommendations, and create personalized looks that reflect their unique fashion preferences.

(198) Some implementations of the virtual styling system feature an extensive database of clothing items, including tops, bottoms, dresses, jackets, and accessories. Users can browse the database, filter items by category, style, color, or brand, and select garments to incorporate into their virtual outfits. Each clothing item may be digitally rendered to accurately represent its design, color, texture, and fit.

(199) In some example implementations, the virtual styling process begins with users selecting clothing items from the database and adding them to their virtual wardrobe. Users can mix and match different garments to create outfits and experiment with various styling combinations.

(200) Some example implementations of the system may employ computer vision and/or machine learning algorithms to analyze the user's body shape, size, and proportions based on images or video input. These algorithms generate a personalized virtual avatar that accurately represents the user's physique. The selected clothing items are dynamically fitted onto the user's virtual avatar, taking into account fabric draping, stretching, and body movement, resulting in a realistic and

visually accurate representation of the outfit.

(201) In an example implementation, the virtual styling system offers a wide range of customization options to users. They can adjust garment parameters such as sleeve length, neckline, waistline, and hemline, enabling them to tailor the clothing items to their desired style. Users can experiment with color variations, patterns, and textures to further customize their outfits.

(202) In some example implementations, the system provides interactive controls that allow users to easily modify the appearance of virtual garments. Users can resize, rotate, and position clothing items on their avatar, ensuring a precise and personalized fit. Furthermore, the system offers suggestions and recommendations based on user preferences, fashion trends, and compatibility between different clothing items.

(203) In some implementation, to enhance the realism of the virtual styling experience, the system employs augmented reality techniques. Users can visualize their styled outfits on their virtual avatar in a realistic virtual environment. Lighting, shadows, and reflections are simulated to accurately represent how the clothing would appear in different settings and under varying conditions.

(204) Users may view their virtual outfits from different angles, zoom in for detailed inspection, and interact with the garments through virtual touch gestures. The system provides a visually immersive experience that allows users to evaluate the styling effects and make informed decisions.

(205) Some implementations may include a virtual styling system utilizes artificial intelligence and machine learning algorithms to offer personalized recommendations to users. By analyzing user preferences, past interactions, and fashion trends, the system can suggest complementary clothing items, accessories, and styling ideas that align with the user's fashion taste and individual style.

(206) In some implementations, the virtual styling system seamlessly integrates with ecommerce platforms, allowing users to directly purchase the clothing items they style or add them to a wishlist for future reference. The system provides links to online retailers, enabling users to access additional product information and make informed purchase decisions. Users may also share their styled outfits on social media platforms, receive feedback from friends.

(207) The present invention relates to a shirt designed with specialized markings to enhance the mapping of clothing between a model's body and a user's body in virtual try-on systems. By incorporating these markings strategically on the shirt, accurate and precise alignment of virtual clothing items on the user's body can be achieved. The shirt enables the virtual try-on system to provide a realistic and visually appealing representation of how clothing would appear on the user. Additionally, the system includes styling capabilities that allow users to personalize and customize their virtual outfits, providing an immersive and engaging virtual shopping experience.

(208) A system in accordance with aspects of the invention leverages computing resources, including machine learning models, rendering engines, and generative AI to provide, in real time, hyper-personalized recommendations for outfits of clothing, handbags, and footwear and, furthermore, coaching to a particular user on how to wear the clothes of the outfits according to a particular fashion trend of interest, optionally in view of the user's human body type.

(209) As used in this specification, the term style generally refers to a fashion trend that may evolve from time to time and include not only patterns of color, fabric, texture, and designs, but also how an article of clothing is cut, shaped, and/or worn. Fashion trends can be referenced by a time period in which they were popular. However, some trends don't evolve and are considered to be classic. One fashion trend, for example, is 2023's summer trend of relaxed, monochrome, and comfortable clothing, which amongst other things includes outfits consisting of crop tops, khakis shorts, and mules. Another example of a fashion trend is the bright colors and rolled up sleeves typically worn in the 80s. But style isn't just about the type, color, material, and design of clothing, it's also about how the clothing is worn, e.g., baggy, tightly, form fitting, collar up, and/or sleeves rolled.

(210) One aspect of hyper-personalization is computer modeling and showing how articles of clothing would actually wear on the particular user in real life. The system includes rendering

models and related computing resources that operate together to generate realistic digital representations of how particular articles of clothing would wear on various human body types. Moreover, the system can individually style these articles, e.g., pushing up sleeves, draping a jacket over shoulders, tucking in or half tucking a top. Hence, multiple representations are generated for each article for both various body types and for various styles. These representations can be stored on network accessible servers. The system selects the appropriate digital representations based on user input specifying the user's human body type, e.g., one or more current photos, videos, standard clothing sizes, and/or detailed measurements typically taken by a tailor. Generally, the system selects the digital representation for the particular user, but optionally can select representations for others as specified by the user (which option is useful, e.g., when the user is shopping for someone else).

(211) Another aspect of hyper-personalization is the system's capacity to remember, learn, and predict not only the user selections of clothes and outfits, e.g., white collared blouses with black jackets, but also the user selections of style, e.g., blouse worn with sleeves rolled over jacket sleeves. The system includes generative AI resources that remembers the user's current and historical preferences, especially with respect to contemporaneous styles, learns current styles, predicts whether the user would be interested in the latter based on the former, and coaches the user on selections of outfits and styles.

(212) The system includes servers and SDKs which can be incorporated into web pages and mobile applications, including those of high-fashion designers and retailers.

(213) The servers include resources that are: (i) machine learning models to predict user preferences based on their previous engagements and the previous engagements of the market segments to which they belong; (ii) classifiers to classify users, articles of clothing, their combinations, and styles based on signals that include historical and current fashion trends; (iii) machine learning models that, given human body types and a particular style of interest, predict how an article of clothing would wear (e.g., how it drapes, where does it stretches, and/or where it creases) on each human body type in the style of interest; (iv) a rendering engine that generates digital representations of articles of clothing, including how they wear on the human body in a particular style; and (v) generative AI resources to learn about historical and current fashion trends and predicts future ones; and (vi) one or more datastores, including logs of user engagements, which are anonymized according to user selection.

(214) The system's SDKs include instructions to enable the system to retrieve information about articles of clothing from designers, retailers, and other B2B customers. The SDKs also include instructions to enable features and functionalities of the system in the websites and mobile applications of B2B customers, examples of which features and functionalities include those called "Complete the Look" and "Zelig Closet," as well as "Zelig Mirror," which is a rendering of the user or a human model, selected by the user, wearing the current articles of clothing tried on during the current session. These features and functionalities are depicted in FIGS. 65A-D, and are described below in reference to these figures.

(215) In operation, the system's server ingest information specifying articles of clothing, e.g., from fashion designers and retailers and uses its computing resources (ML models, GenAI resources, and rendering engine) to process this information to generate and store digital representations of the articles, which when rendered depicts ultra-realistically how they would appear on human body types.

(216) During a user browsing session on the designer or retailer's website or mobile application (or other computing clients of other B2B customers), the system's SDKs, which include computing instructions, work in conjunction with the system's servers to select the human body type best suited for the particular user browsing, based on a photo or video or other information submitted by the user, including user selection, which is depicted in FIG. 65A. The system's servers then send the appropriate digital representations to the websites or mobile applications or other clients for

rendering, usually in response to user selection to try on the corresponding articles of clothing and style (e.g., how the article is to be worn). Moreover, servers and clients work in conjunction to provide styling guidance and recommendations to the user based on their profiles and engagements, current and historical.

(217) In response to a user selection of a call to action called “Complete the Look,” as shown in FIGS. 65B and 65C, the system selects additional articles of clothing to recommend in accordance with the style, optionally by leveraging its AI and other computing resources to infer the style preferred by the user in the current session. The system can automatically update a rendering of the user with the additional articles of clothing and in the inferred style.

(218) The rendering of the user wearing articles of clothing of interest in the style of interest functions as a mirror through which the user can see, in real time, what they would look like in real life. This feature is called the “Zelig Mirror,” examples of which are depicted in FIGS. 65B-D and 66. Notably Zelig Mirror can appear as its own window on a display and can follow the user as they browse through various pages of a website or mobile application. In this sense Zelig Mirror is the digital analog of having a portable changing room with a mirror as one wanders throughout a brick and mortar store trying on clothes.

(219) Zelig Mirror functions across the system's various modes, e.g., when the user is manually selecting outfits and styles and when the user has opted to get recommendations and guidance for clothing selection and/or style options from the system. In the latter mode, the system coaches the user through their decision making and helps them keep the look going (e.g., letting the user know whether a selection of clothes or style is consistent with a fashion trend of interest).

(220) The user can take a snapshot of Zelig Mirror to store instances of it in Zelig Closet. (An example of this call to action button is shown in FIGS. 65B and C and 66.) Thereby, the user can recall every combination of clothing and every outfit tried on. The user can open their Zelig Closet at any time, especially during browsing sessions when they have tried on too many outfits to keep in mind. Zelig Closet advantageously facilitates decision making by presenting side-by-side comparison of various outfits under consideration.

(221) The system can be implemented as a game and/or as a B2B service for retailers and designers. It can also be implemented as a fashion coaching tool for education, not only for end users, but also for individual stylists and in-store sales associates. And it can be implemented as a service personalized for the user. In the latter case, the system can generate a digital representation of the user's particular human body (rather than match the user to one of its human body types).

(222) Optionally, the system aggregates and where appropriate anonymizes session information and provides a dashboard of statistics to B2B customers, including reduction in merchandise rate of return and/or exchange that is enabled by the system and the corresponding savings of monetary, energy savings, and other resources. Optionally included on the dashboard are statistics of user engagements and inferences and insights gained from them, where user consent has been received. This information can enhance the personalization of recommendations.

(223) One or more elements or aspects or steps, or any portion(s) thereof, from one or more of any of the systems and methods described herein may be combined with one or more elements or aspects or steps, or any portion(s) thereof, from one or more of any of the other systems and methods described herein and combinations thereof, to form one or more additional implementations and/or claims of the present disclosure.

(224) One or more of the components, steps, features, and/or functions illustrated in the figures may be rearranged and/or combined into a single component, block, feature or function or embodied in several components, steps, or functions. Additional elements, components, steps, and/or functions may also be added without departing from the disclosure. The apparatus, devices, and/or components illustrated in the Figures may be configured to perform one or more of the methods, features, or steps described in the Figures. The algorithms described herein may also be efficiently implemented in software and/or embedded in hardware.

(225) Reference in the specification to “one implementation” or “an implementation” means that a particular feature, structure, or characteristic described in connection with the implementation is included in at least one implementation of the invention. The appearances of the phrase “in one implementation” in various places in the specification are not necessarily all referring to the same implementation.

(226) Some portions of the detailed description are presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the methods used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared or otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers or the like.

(227) It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the following disclosure, it is appreciated that throughout the disclosure terms such as “processing,” “computing,” “calculating,” “determining,” “displaying” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system's memories or registers or other such information storage, transmission or display.

(228) Finally, the algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general-purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct more specialized apparatus to perform the required method steps. The required structure for a variety of these systems will appear from the description below. It will be appreciated that a variety of programming languages may be used to implement the teachings of the invention as described herein.

(229) The figures and the following description describe certain implementations by way of illustration only. One skilled in the art will readily recognize from the following description that alternative implementations of the structures and methods illustrated herein may be employed without departing from the principles described herein. Reference will now be made in detail to several implementations, examples of which are illustrated in the accompanying figures. It is noted that wherever practicable similar or like reference numbers may be used in the figures to indicate similar or like functionality.

(230) The foregoing description of the implementations of the present invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the present invention be limited not by this detailed description, but rather by the claims of this application. As will be understood by those familiar with the art, the present invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. Likewise, the particular naming and division of the modules, routines, features, attributes, methodologies and other aspects are not mandatory or significant, and the mechanisms that implement the present invention or its features may have different names, divisions and/or formats.

(231) Furthermore, as will be apparent to one of ordinary skill in the relevant art, the modules, routines, features, attributes, methodologies and other aspects of the present invention can be

implemented as software, hardware, firmware or any combination of the three. Also, wherever a component, an example of which is a module, of the present invention is implemented as software, the component can be implemented as a standalone program, as part of a larger program, as a plurality of separate programs, as a statically or dynamically linked library, as a kernel loadable module, as a device driver, and/or in every and any other way known now or in the future to those of ordinary skill in the art of computer programming.

(232) Additionally, the present invention is in no way limited to implementation in any specific programming language, or for any specific operating system or environment. Accordingly, the disclosure of the present invention is intended to be illustrative, but not limiting, of the scope of the present invention, which is set forth in the following claims.

(233) It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order and are not meant to be limited to the specific order or hierarchy presented.

(234) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term “some” refers to one or more. Combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words “module,” “mechanism,” “element,” “device,” and the like may not be a substitute for the word “means.” As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.”

Claims

1. A method comprising: ingesting, using a server, information specifying articles of clothing using a first machine learning model to generate a database of garments, each garment in the database having a plurality of labels, at least one label being one or more styling combinations; receiving, from a user, selection of a first body type; determining a first garment based on user interaction with a user interface of a platform for purchasing clothing having a virtual try-on capability; determining, using a server, an inferred style based upon the first garment with a second machine learning model, wherein the inferred style includes at least one styling combination; retrieving,

from the database of garments, one or more additional garments, a footwear, and an accessory based upon the inferred style and the first garment and combining the one or more additional garments, the footwear, the accessory and the first garment to create a stored outfit with a look, wherein the look is garments, footwear, and an accessory stylized to a selected preference of the user in the at least one styling combination; generating a shopper avatar having a second body type matching the first body type and wearing the stored outfit with the inferred style showing the look, wherein the first garment and the one or more additional garments are warped and aligned using a third machine learning model to deform the first garment and the one or more additional garments over the shopper avatar including showing a clothing interaction of the first garment with the one or more additional garments; and providing the shopper avatar wearing the stored outfit with the inferred style showing the look including the first garment and the one or more additional garments are warped and aligned and the clothing interaction for display in a user interface in real-time so that the user interface serves as a mirror through which the user can see what they would look like in real life.

2. The method of claim 1, wherein the styling combination includes one or more from a group of a fashion standard styling segment, a top styling segment having one of from a group of a following styling segments including sleeves buttoned, half tucked, fully buttoned front portion and collar folded, sleeve length styling segment having potential length values of wrist, mid hand and fingers, and bodice fit styling segment having potential fit values of tight, relaxed and oversized.

3. The method of claim 1, further comprising: receiving a selection of a new garment by the user, the selection initiating a virtual try-on of the new garment to replace the first garment with the new garment based on the new garment and the first garment having a same type of clothing; configuring the shopper avatar to wear the new garment according to one or more styling options; and providing the shopper avatar wearing the new garment for display to the user in the user interface.

4. The method of claim 1, further comprising: selecting an additional garment of clothing to recommend to the user using the second machine learning model in accordance with the inferred style of the user; updating the user interface to include the additional garment in the inferred style along with the shopper avatar; and providing the updating the user interface for display to the user.

5. The method of claim 1, wherein the information specifying articles of clothing is from one or more from a group of: a designer, a retailer, a B2B customer, a website, a B2B application, and a mobile application.

6. The method of claim 1, wherein ingesting information specifying articles of clothing using the first machine learning model is performed using artificial intelligence and machine learning algorithms and a rendering engine.

7. The method of claim 1, wherein ingesting information specifying articles of clothing using the first machine learning model, further comprises: selecting a subset of a large dataset of images of garments from a pre-existing database, wherein the subset of the large data set of images are labelled with a specific style; generating a digital taxonomy having one or more labels associated with each image in the subset of the large dataset of images, wherein at least one of the one or more labels corresponds to the specific style; training a style classifier using the digital taxonomy applied to the subset of the large dataset of images of garments; and generating a first dataset based on executing the trained style classifier on the large dataset of images of garments, the first dataset comprising a plurality of images of garments meeting a threshold probability of depicting the specific style.

8. The method of claim 1, wherein the shopper avatar is a rendering of the user or a human model.

9. The method of claim 1, wherein receiving, from a user, selection of a first body type further comprises: presenting a user interface with a plurality of human body types; receiving a selection of one of the plurality of human body types; and presenting the selection of the one of the plurality of human body types as the first body type for the shopper avatar in the user interface along with

body dimensions.

10. The method of claim 9, wherein receiving the selection of one of the plurality of human body types comprises: receiving a photograph or a video of the user; analyzing the photograph or video of the user; and selecting one of the plurality of human body types as the first body type based on the analyzing the photograph or video of the user.

11. The method of claim 1, wherein the second machine learning model infers the inferred style for the stored outfit preferred by the user based upon styling guidance and recommendations for the user based on a current and historical profile and current and historical engagements of the user.

12. The method of claim 1, wherein the user interface in a window overlaid on a web page of a web browser showing in the web page of a website of a retailer.

13. The method of claim 1, wherein the warping and aligning the one garment over the shopper avatar uses a warping model and a synthesis model.

14. A system comprising one or more processors and memory operably coupled with the one or more processors, wherein the memory stores instructions that, in response to execution of the instructions by the one or more processors, cause the one or more processors to perform the following operations: ingesting information specifying articles of clothing using a first machine learning model to generate a database of garments, each garment in the database having a plurality of labels, at least one label being one or more styling combinations; receiving, from a user, selection of a first body type; determining a first garment based on user interaction with a user interface of a platform for purchasing clothing having a virtual try-on capability; determining an inferred style based upon the first garment with a second machine learning model, wherein the inferred style includes at least one styling combination; retrieving, from the database of garments, one or more additional garments, a footwear, and an accessory based upon the inferred style and the first garment and combining the one or more additional garments, the footwear, the accessory and the first garment to create a stored outfit with a look, wherein the look is garments, footwear, and an accessory stylized to a selected preference of the user in the at least one styling combination; generating a shopper avatar having a second body type matching the first body type and wearing the stored outfit with the inferred style showing the look, wherein the first garment and the one or more additional garments are warped and aligned using a third machine learning model to deform the first garment and the one or more additional garments over the shopper avatar including showing a clothing interaction of the first garment with the one or more additional garments; and providing the shopper avatar wearing the stored outfit with the inferred style showing the look including the first garment and the one or more additional garments are warped and aligned and the clothing interaction for display in a user interface in real-time so that the user interface serves as a mirror through which the user can see what they would look like in real life.

15. The system of claim 14, wherein the styling combination includes one or more from a group of a fashion standard styling segment, a top styling segment having one of from a group of a following styling segments including sleeves buttoned, half tucked, fully buttoned front portion and collar folded, sleeve length styling segment having potential length values of wrist, mid hand and fingers, and bodice fit styling segment having potential fit values of tight, relaxed and oversized.

16. The system of claim 14, wherein the operations further comprise: receiving a selection of a new garment by the user, the selection initiating a virtual try-on of the new garment to replace the first garment with the new garment based on the new garment and the first garment having a same type of clothing; configuring the shopper avatar to wear the new garment according to one or more styling options; and providing the shopper avatar wearing the new garment for display to the user in the user interface.

17. The system of claim 14, wherein the operations further comprise: selecting an additional garment of clothing to recommend to the user using the second machine learning model in accordance with the inferred style of the user; updating the user interface to include the additional garment in the inferred style along with the shopper avatar; and providing the updating the user

interface for display to the user.

18. The system of claim 14, wherein the information specifying articles of clothing is from one or more from a group of: a designer, a retailer, a B2B customer, a website, a B2B application, and a mobile application.

19. The system of claim 14, wherein ingesting information specifying articles of clothing using the first machine learning model is performed using artificial intelligence and machine learning algorithms and a rendering engine.

20. The system of claim 14, wherein ingesting information specifying articles of clothing using the first machine learning model further comprises the operations of: selecting a subset of a large dataset of images of garments from a pre-existing database, wherein the subset of the large data set of images are labelled with a specific style; generating a digital taxonomy having one or more labels associated with each image in the subset of the large dataset of images, wherein at least one of the one or more labels corresponds to the specific style; training a style classifier using the digital taxonomy applied to the subset of the large dataset of images of garments; and generating a first dataset based on executing the trained style classifier on the large dataset of images of garments, the first dataset comprising a plurality of images of garments meeting a threshold probability of depicting the specific style.

21. The system of claim 14, wherein the shopper avatar is a rendering of the user or a human model.

22. The system of claim 14, wherein receiving, from a user, selection of a first body type further comprises: presenting a user interface with a plurality of human body types; receiving a selection of one of the plurality of human body types; and presenting the selection of the one of the plurality of human body types as the first body type for the shopper avatar in the user interface along with body dimensions.

23. The system of claim 22, wherein receiving the selection of one of the plurality of human body types comprises: receiving a photograph or a video of the user; analyzing the photograph or video of the user; and selecting one of the plurality of human body types as the first body type based on the analyzing the photograph or video of the user.

24. The system of claim 14, wherein the second machine learning model infers the style for the stored outfit preferred by the user based upon styling guidance and recommendations for the user based on a current and historical profile and current and historical engagements of the user.

25. The system of claim 14, wherein the user interface in a window overlaid on a web page of a web browser showing in the web page of a website of a retailer.

26. The system of claim 14, wherein the warping and aligning the one garment over the shopper avatar uses a warping model and a synthesis model.
